The effects of blood pressure reduction and of different blood pressure-lowering regimens on major cardiovascular events according to baseline bloo...

BACKGROUND: The benefits of reducing blood pressure are well established, but there remains uncertainty about whether the magnitude of the effect varies with the initial blood pressure level. The objective was to compare the risk reductions achieved by different blood pressure-lowering regimens among individuals with different baseline blood pressures. METHODS: Thirty-two randomized controlled trials were included and seven comparisons between different types of treatments were made. For each comparison, the primary prespecified analysis included calculation of summary estimates of effect using random-effects meta-analysis for major cardiovascular events in four groups defined by baseline SBP (<140, 140-159, 160-179, and ≥ 180 mmHg). RESULTS: There were 201 566 participants among whom 20 079 primary outcome events were observed. There was no evidence of differences in the proportionate risk reductions achieved with different blood pressure-lowering regimens across groups defined according to higher or lower levels of baseline SBP (all P for trend > 0.17). This finding was broadly consistent for comparisons of different regimens, for DBP categories, and for commonly used blood pressure cut-points. CONCLUSION: It appears unlikely that the effectiveness of blood pressure-lowering treatments depends substantively upon starting blood pressure level. As the majority of patients in the trials contributing to these overviews had a history of hypertension or were receiving background blood pressure-lowering therapy, the findings suggest that additional blood pressure reduction in hypertensive patients meeting initial blood pressure targets will produce further benefits. More broadly, the data are supportive of the utilization of blood pressure-lowering regimens in high-risk patients with and without hypertension.